

13.1 - Fibonacci Numbers

Problem Solving, Math 1000

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1 Definition

Beginning with $F_1 = 1$ and $F_2 = 1$, the Fibonacci numbers are defined by $F_N = F_{N-1} + F_{N-2}$. This sequence of numbers, 1, 1, 2, 3, 4, 8, 13, 21, 34, 55, 89, 144, ... is known as the Fibonacci sequence, and has many interesting properties.

2 Question

Notice that the Fibonacci numbers are defined recursively. With this in mind, one might observe that calculating F_N for a large N might take quite awhile if this is our only option (i.e. calculating F_{100} this way requires calculating all the Fibonacci numbers before it). To remedy this problem, some mathematicians found a closed-form formula for the N th Fibonacci number.

3 Theorem (Binet's Formula)

The N th Fibonacci number is given by

$$F_N = \left[\left(\frac{1 + \sqrt{5}}{2} \right)^N - \left(\frac{1 - \sqrt{5}}{2} \right)^N \right] / \sqrt{5}$$

To make this even easier, notice that we may disregard the second term in the numerator and simply rounding to the nearest integer. That is, calculate

$$F_N \approx \left(\frac{1 + \sqrt{5}}{2} \right)^N / \sqrt{5}$$

and round to the nearest integer to find F_N .